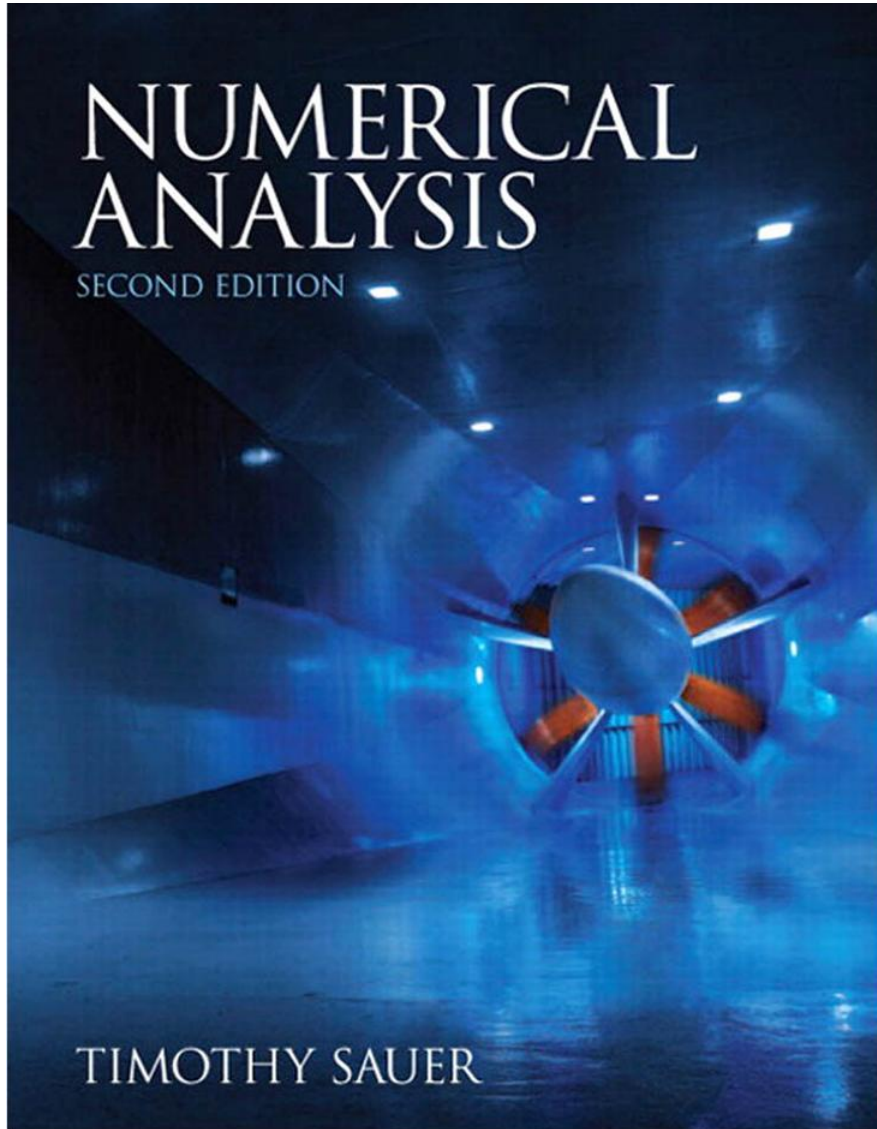


NUMERICAL ANALYSIS

SECOND EDITION

TIMOTHY SAUER



Root Finding Methods

$f(x) = 0$ [polynomial equation]

$$ax + b = 0$$

$$ax^2 + bx + c = 0$$

$$ax^3 + bx^2 + cx + d = 0$$



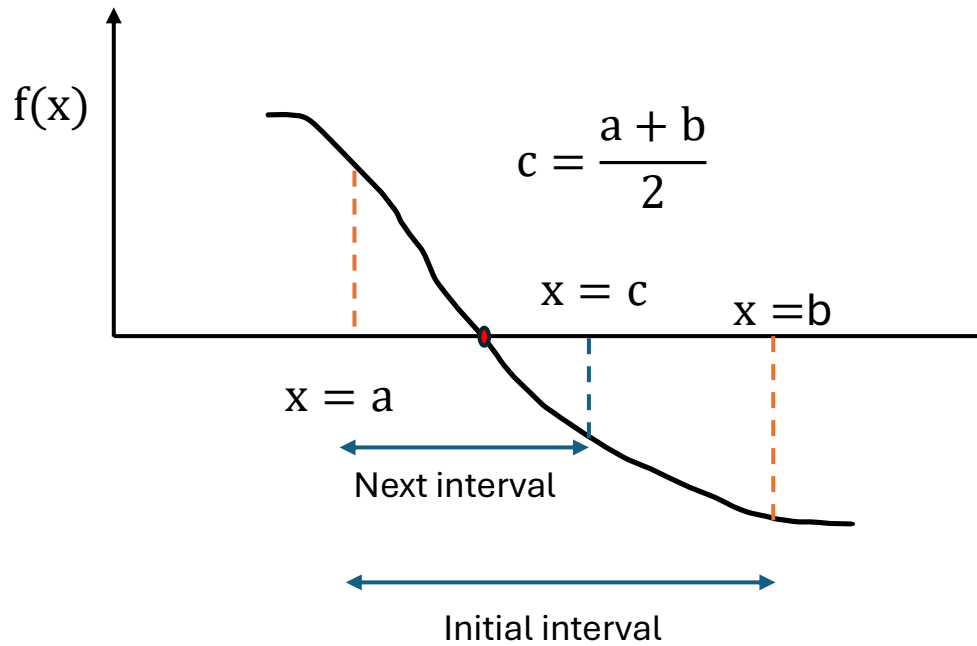
Geralamo Cardano



Fontana Tartaglia

Bisection Method:

$$f(x) = 0$$



Pseudocode:

```
Given initial interval  $[a, b]$  such that  $f(a)f(b) < 0$   
while  $(b - a)/2 > \text{TOL}$   
     $c = (a + b)/2$   
    if  $f(c) = 0$ , stop, end  
    if  $f(a)f(c) < 0$   
         $b = c$   
    else  
         $a = c$   
    end  
end  
The final interval  $[a, b]$  contains a root.  
The approximate root is  $(a + b)/2$ .
```

Example:

Find a root of the function $f(x) = x^3 + x - 1$ by using the Bisection Method on the interval $[0, 1]$.

i	a_i	$f(a_i)$	c_i	$f(c_i)$	b_i	$f(b_i)$
0	0.0000	—	0.5000	—	1.0000	+
1	0.5000	—	0.7500	+	1.0000	+
2	0.5000	—	0.6250	—	0.7500	+
3	0.6250	—	0.6875	+	0.7500	+
4	0.6250	—	0.6562	—	0.6875	+
5	0.6562	—	0.6719	—	0.6875	+
6	0.6719	—	0.6797	—	0.6875	+
7	0.6797	—	0.6836	+	0.6875	+
8	0.6797	—	0.6816	—	0.6836	+
9	0.6816	—	0.6826	+	0.6836	+

root, $r = [0.6816, 0.6826] \approx 0.6821$ (mid value)

Newton-Raphson Method:

$$f(x) = 0$$

Take an initial guess x_0

$$f(x) = f(x_0) + (x - x_0)f'(x_0)$$

$$0 = f(x_0) + (x - x_0)f'(x_0)$$

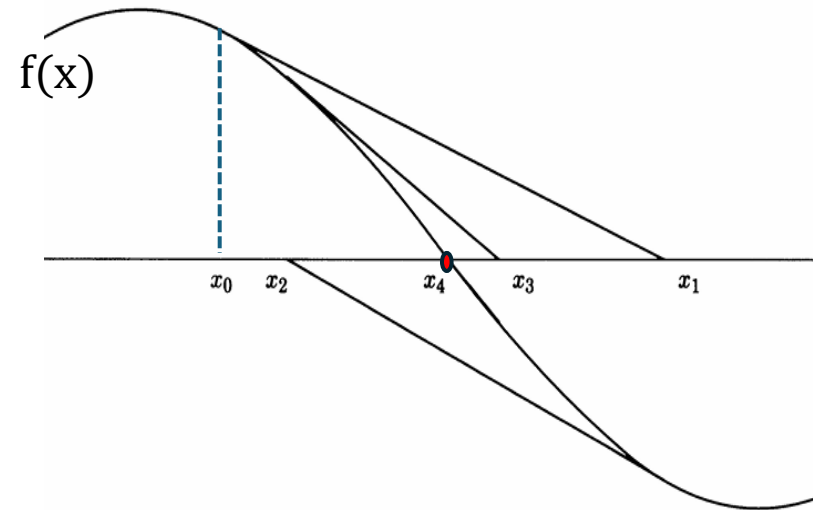
$$x = x_0 - f(x_0)/f'(x_0)$$

Geometrical Interpretation:

$$f'(x_0)(x_1 - x_0) = 0 - f(x_0)$$

$$x_1 - x_0 = -\frac{f(x_0)}{f'(x_0)}$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}.$$



$$f(x) = x^3 + x - 1$$

Bisection Method:

i	a_i	$f(a_i)$	c_i	$f(c_i)$	b_i	$f(b_i)$
0	0.0000	—	0.5000	—	1.0000	+
1	0.5000	—	0.7500	+	1.0000	+
2	0.5000	—	0.6250	—	0.7500	+
3	0.6250	—	0.6875	+	0.7500	+
4	0.6250	—	0.6562	—	0.6875	+
5	0.6562	—	0.6719	—	0.6875	+
6	0.6719	—	0.6797	—	0.6875	+
7	0.6797	—	0.6836	+	0.6875	+
8	0.6797	—	0.6816	—	0.6836	+
9	0.6816	—	0.6826	+	0.6836	+

Newton-Raphson Method:

i	x_i
0	−0.70000000
1	0.12712551
2	0.95767812
3	0.73482779
4	0.68459177
5	0.68233217
6	0.68232780
7	0.68232780

root, $r = [0.6816, 0.6826] \approx 0.6821$ (mid value)

Error and Convergence:

Bisection Method:

$$\text{Error} = (b-a)/2^n$$

Newton-Raphson Method:

$$f(r) = f(x_i) + (r - x_i)f'(x_i) + \frac{(r - x_i)^2}{2}f''(c_i).$$

$$0 = f(x_i) + (r - x_i)f'(x_i) + \frac{(r - x_i)^2}{2}f''(c_i)$$

$$\frac{f(x_i)}{f'(x_i)} = r - x_i + \frac{(r - x_i)^2}{2} \frac{f''(c_i)}{f'(x_i)},$$

$$x_i - \frac{f(x_i)}{f'(x_i)} - r = \frac{(r - x_i)^2}{2} \frac{f''(c_i)}{f'(x_i)}$$

$$x_{i+1} - r = e_i^2 \frac{f''(c_i)}{2f'(x_i)}$$

$$e_{i+1} = e_i^2 \left| \frac{f''(c_i)}{2f'(x_i)} \right| \qquad \lim_{i \rightarrow \infty} \frac{e_{i+1}}{e_i^2} = \left| \frac{f''(r)}{2f'(r)} \right|,$$